

Pedestrian and Bicycle Infrastructure: A National Study of Employment Impacts

Garrett-Peltier, Heidi | 6/20/2011

Abstract:

Pedestrian and bicycling infrastructure such as sidewalks, bike lanes, and trails, can all be used for transportation, recreation, and fitness. These types of infrastructure have been shown to create many benefits for their users as well as the rest of the community. Some of these benefits are economic, such as increased revenues and jobs for local businesses, and some are non-economic benefits such as reduced congestion, better air quality, safer travel routes, and improved health outcomes. While other studies have examined the economic and non-economic impacts of the use of walking and cycling infrastructure, few have analyzed the employment that results from the design and construction of these projects. In this study we estimate the employment impacts of building and refurbishing transportation infrastructure for cyclists and pedestrians. We analyze various transportation projects and use state-specific data to estimate the number of jobs created within each state where the project is located.

The data for this study were gathered from departments of transportation and public works departments from 11 cities in the United States. Using detailed cost estimates on a variety of projects, we use an input-output model to study the direct, indirect, and induced employment that is created through the design, construction, and materials procurement of bicycle, pedestrian, and road infrastructure. We evaluate 58 separate projects and present the results by project, by city, and by category. Overall we find that bicycling infrastructure creates the most jobs for a given level of spending: For each \$1 million, the cycling projects in this study create a total of 11.4 jobs within the state where the project is located. Pedestrian-only projects create an average of about 10 jobs per \$1 million and multi-use trails create nearly as many, at 9.6 jobs per \$1 million. Infrastructure that combines road construction with pedestrian and bicycle facilities creates slightly fewer jobs for the same amount of spending, and road-only projects create the least, with a total of 7.8 jobs per \$1 million. On average, the 58 projects we studied create about 9 jobs per \$1 million within their own states. If we add the spill-over employment that is created in other states through the supply chain, the employment impact rises by an average of 3 additional jobs per \$1 million.

http://www.peri.umass.edu/fileadmin/pdf/published_study/PERI_ABikes_June2011.pdf